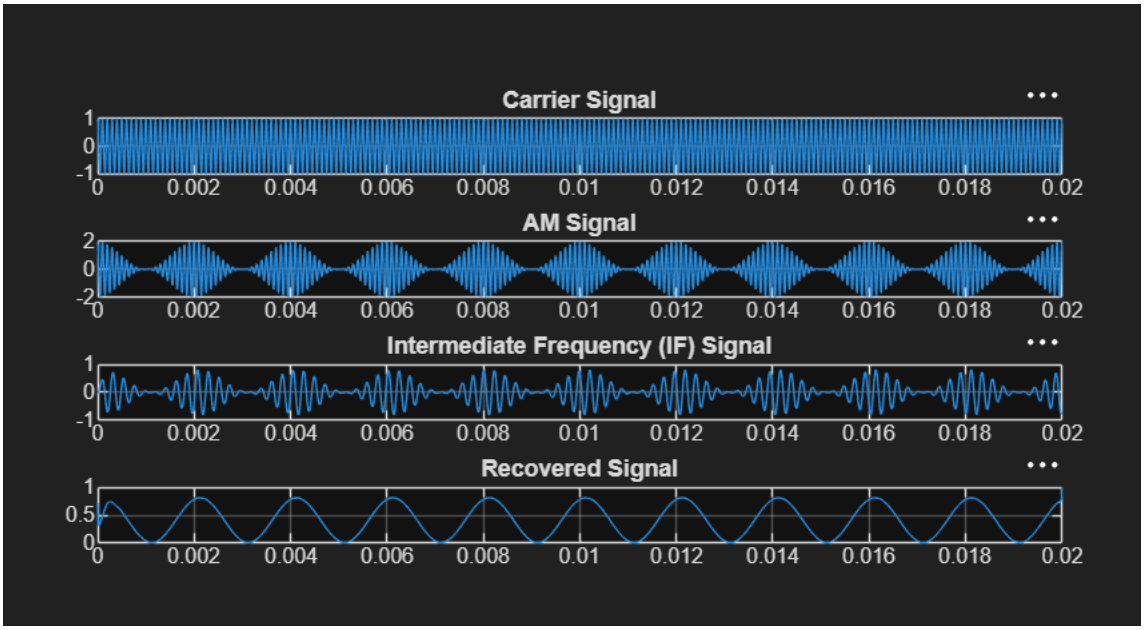
MATLAB SIMULATION OF SUPERHETERODYNE RECEIVER FOR AM COMMUNICATION SYSTEM WITH FREQUENCY MIXING AND FILTERING

```
clc; clear;
close all;
% Parameters fs = 100000; t = 0:1/fs:0.02; fm =
500; % Message frequency fc = 10000; %
Carrier frequency fLO = 14550; % Local
Oscillator (IF = 4550 Hz)
% Message Signal


m = cos(2*pi*fm*t); %
Carrier Signal c =
cos(2*pi*fc*t); % AM Signal
am = (1 + m).*cos(2*pi*fc*t);
% Local Oscillator
LO = cos(2*pi*fLO*t); %
Frequency Mixing mixed
= am .* LO;
% IF Filter
[b,a] = butter(4,5000/(fs/2));
IF = filter(b,a,mixed); % Envelope
Detection env = abs(hilbert(IF)); %
Display figure; subplot(4,1,1) plot(t,m)
title('Message Signal') grid on figure;
```

```
subplot(5,1,2) plot(t,c) title('Carrier
Signal') grid on subplot(5,1,3) plot(t,am)
title('AM Signal') grid on subplot(5,1,4)
plot(t,IF) title('Intermediate Frequency
(IF) Signal') grid on subplot(5,1,5)
plot(t,env) title('Recovered Signal')
grid on
```